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Setting up NeuroElectrics Enobio 2

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Setting up NeuroElectrics Enobio

How to setup NeuroElectrics Enobio EEG systems with iMotions.

Enobio EEG systems connect to iMotions through the NeuroElectrics Instrument Control (NIC) software and Lab Streaming Layer (LSL).

Overview

To setup your Enobio with iMotions:

1. Install Neuroelectrics NIC software (only for initial setup).
2. Connect your Enobio to the NIC software via USB, WiFi, or Bluetooth.
3. Connect your Enobio to iMotions via Lab Streaming Layer (LSL) from NIC software.

Notes about using Enobio with iMotions:

- Compatible with any Enobio Headset (8, 20, or 32 channel systems).
- Available in iMotions v9.1.6 or newer.
    - iMotions versions 7.0 to 9.1.5 do not include the device profile settings.
    - If you collected Enobio data in iMotions 9.1.5 and updated to 9.1.6, you may need to re-run the Power Spectral Density (R Notebook) for PSD data and Frontal Asymmetry Analysis (R Notebook) for FAA data.

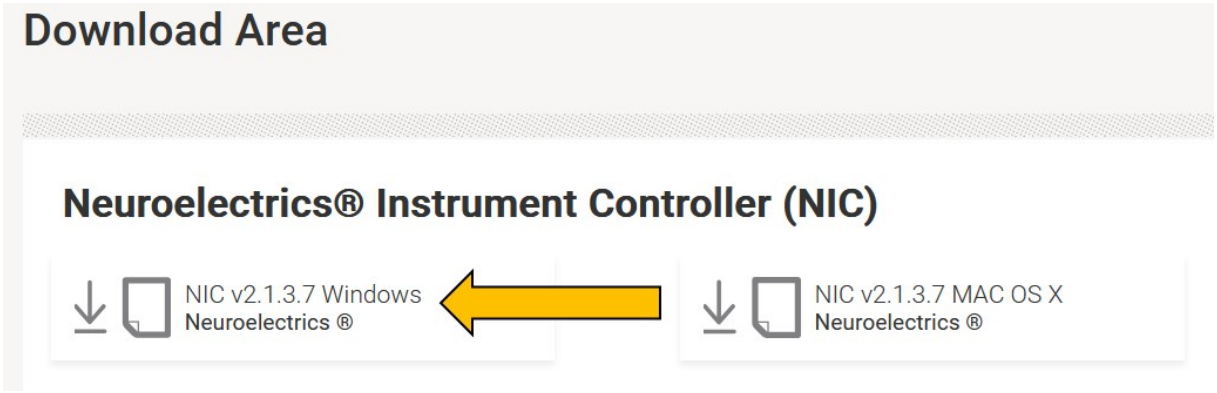
Connecting Enobio to NIC

Download and Install NIC

- NeuroElectrics Instrument Control (NIC) software must be installed and opened.
- Download the latest NIC Windows software from the NeuroElectrics website - external links:
    - NIC for Windows (NIC Download Area)
    - Enobio Manuals

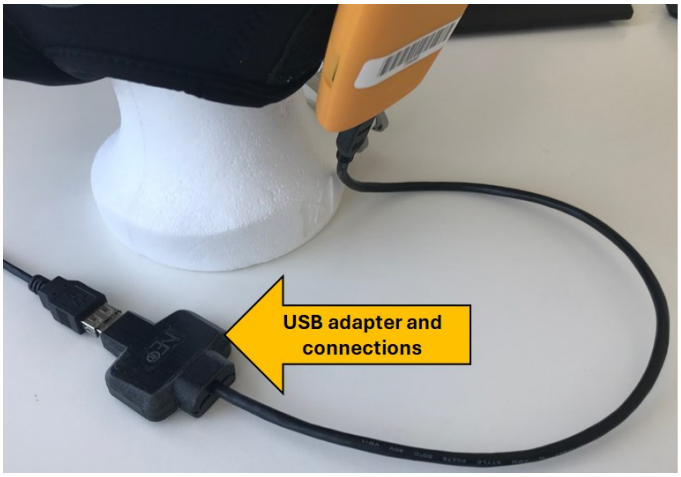
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- Note: Newer version may be available (and are generally recommended).



The Enobio EEG headset can be connected to your computer via USB cable (recommended), WiFi, or BlueTooth.

- USB connection requires the USB adapter (provided by NeuroElectrics with your Enobio) to connect the headset into a USB 3.0 port of the computer.
- WiFi and Bluetooth connections must be connected in Windows WiFi and Bluetooth menus/settings.
- Note: Older Enobio headsets may only support Bluetooth instead of WiFi.





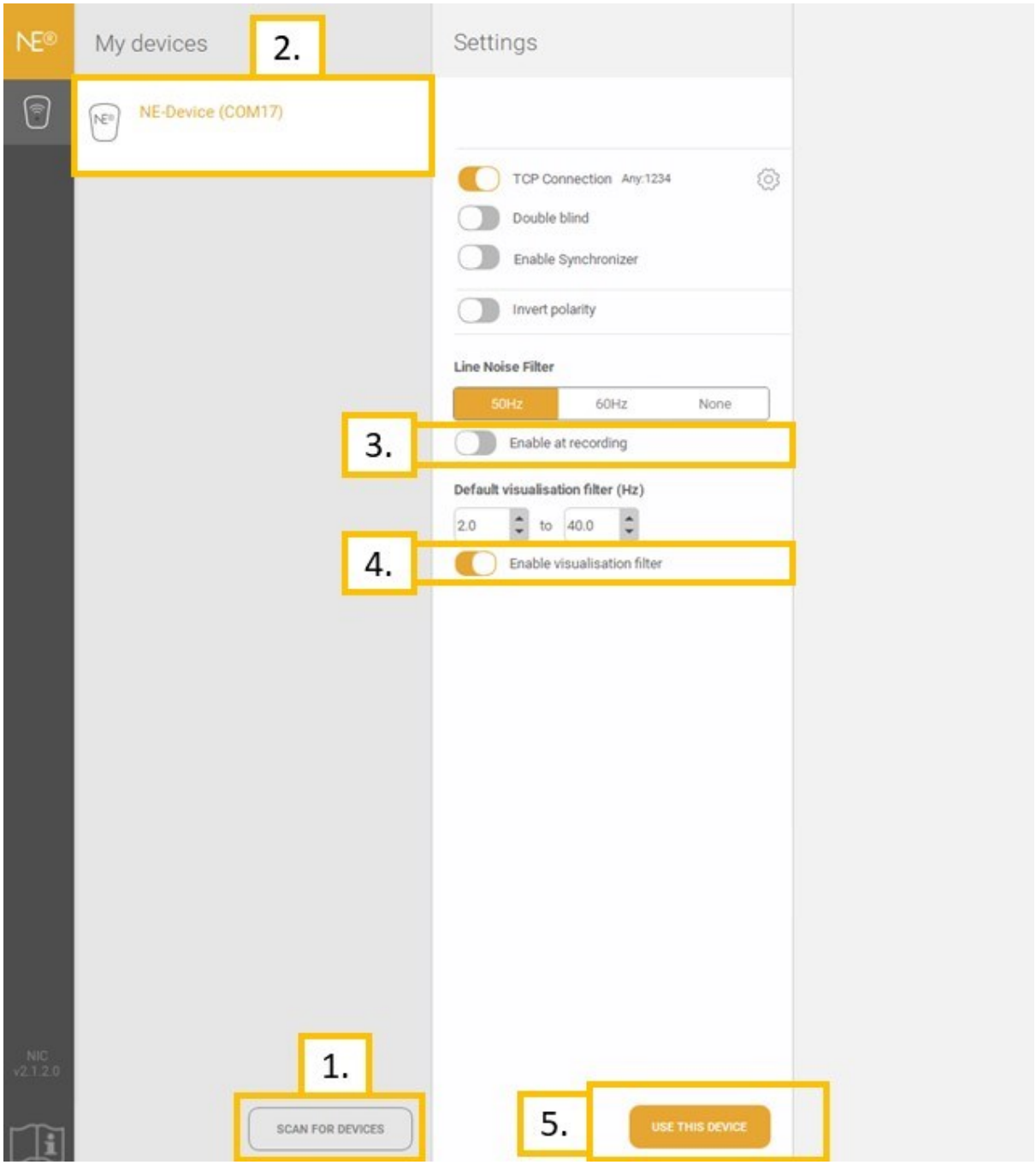
Connect the Enobio in NIC

1. Open the NIC software and click 'SCAN FOR DEVICES' and select your connection type (USB, Wi-Fi, or Bluetooth).
2. Click on your device and enable 'TCP Connection'

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a. Note the device name and COM number, which also should appear in iMotions settings, i.e. 'NE-Device (COM17)' in the screenshot example.
b. The COM port number corresponds with specific USB ports (and will change when different ports are used).

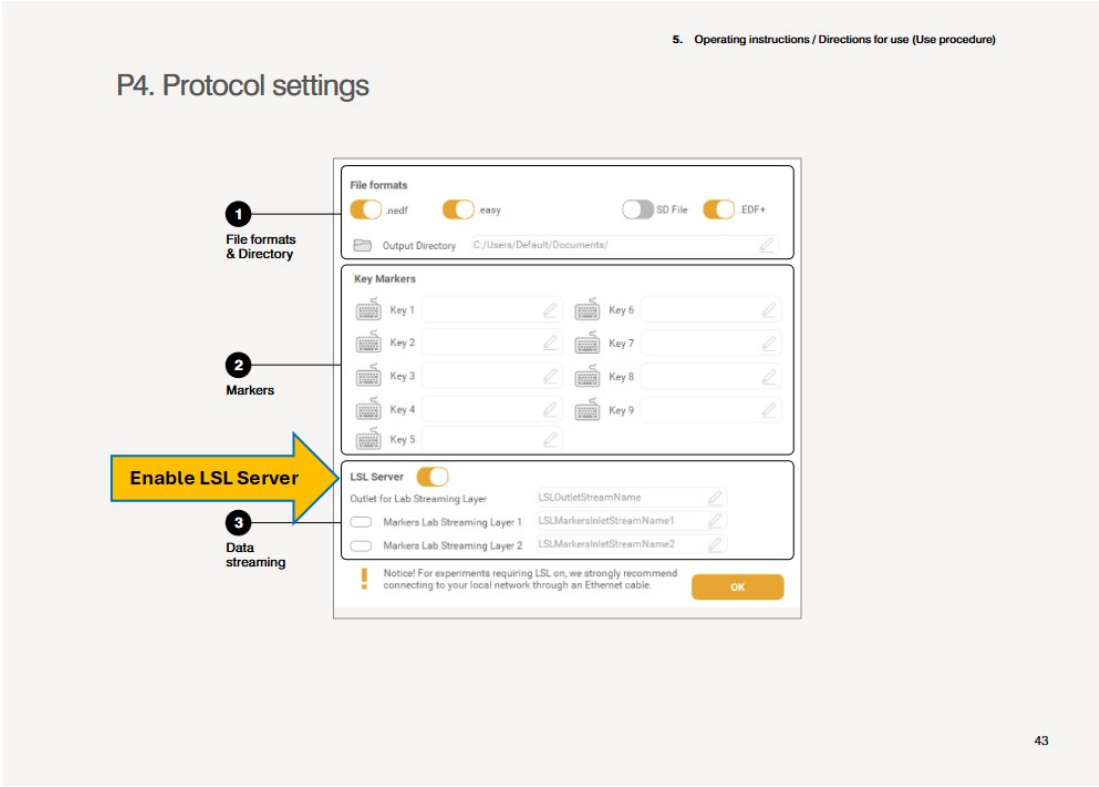
3. You can disable the setting for the Line Noise Filter - Enable at recording (a line noise filter is applied by iMotions EEG Signal Processing).
4. Visualization Filters can remain on default settings and only affect the appearance of data in NIC.
5. Click 'USE THIS DEVICE'. There is a head icon on the left side of the interface.



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Configure and Load Enobio 'Protocol'

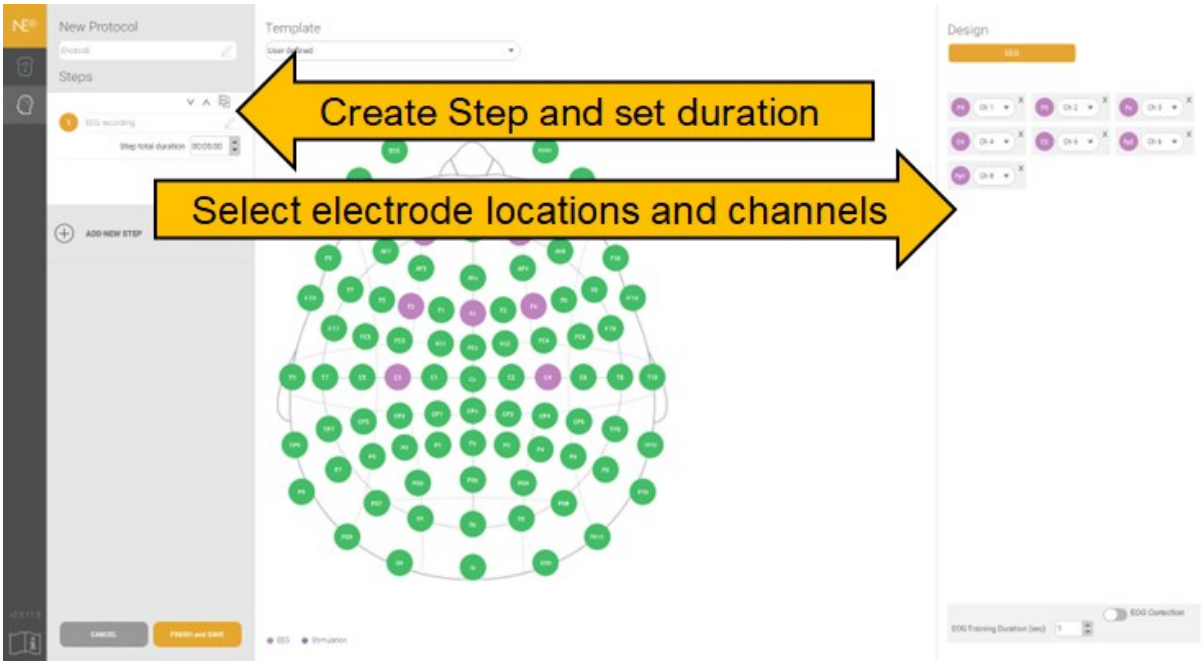
1. To open the NIC Protocols page, click the button at the top-left (head/electrode icon)
2. Click "ADD NEW PROTOCOL" or select an existing protocol (when applicable).
3. Enter a protocol name (e.g., Enobio8).
4. Under "Steps", enter a name (e.g., 8 channels) and an arbitrary step size (here, 20 seconds).
5. Then drag and drop the desired EEG electrodes from the montage onto the right side of the dialog. Make sure that the channels are placed in the correct sequence (You have to at least select one electrode in order for NIC to start a protocol. LSL Outlet will always have the 8/20/32 channels no matter what you setup in NIC).
6. Click "FINISH and SAVE".
7. Click "LOAD PROTOCOL" to start streaming data. NIC protocol must be loaded/running to send data to iMotions.
8. Return to Protocol and open Settings - confirm that LSL Server is enabled in NIC.



9. NIC Liveview page(signal/wave icon) will now show data from the Enobio
    a. The Liveview page is used to assess signals from each EEG electrode

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b. Signal Quality status is shown for each electrode - indicated by the Red/Yellow/Green status for each electrode.
c. Note: If you are *not* using the ear clip, you must place the CMS and DRL leads as close together as possible without them touching to get better impedance values.



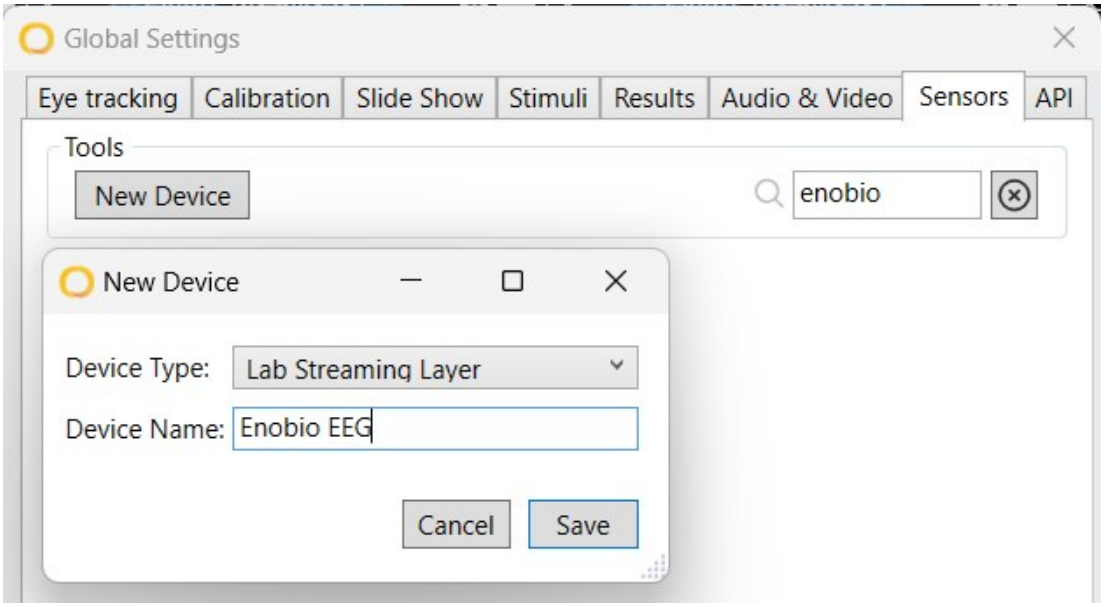
Connect the Enobio to iMotions

The initial connection between your Enobio and iMotion will require set up in your iMotions Global Setings - Sensors tab (New Device process). Once this is done, you will see the Enobio widget in the main iMotions windows and can create a study to record from the Lab Streaming Layer and Enobio.

Adding a new Enobio device

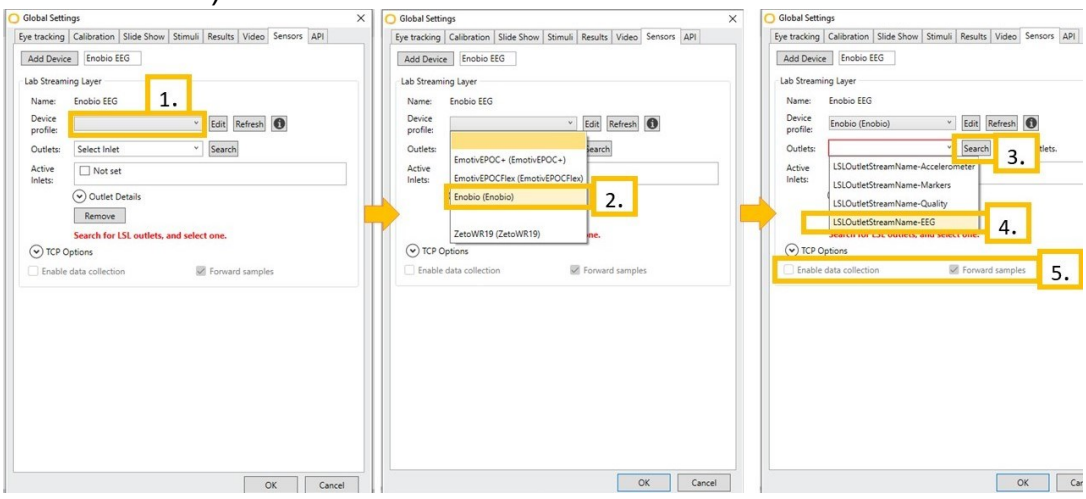
1. In iMotions, open Global Settings - Sensors tab.
2. Click on New Device.
3. Select the Device Type "Lab Streaming Layer"
4. Enter a device name (e.g. Enobio EEG) and click "Save".

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Connect Enobio Device in iMotions Settings

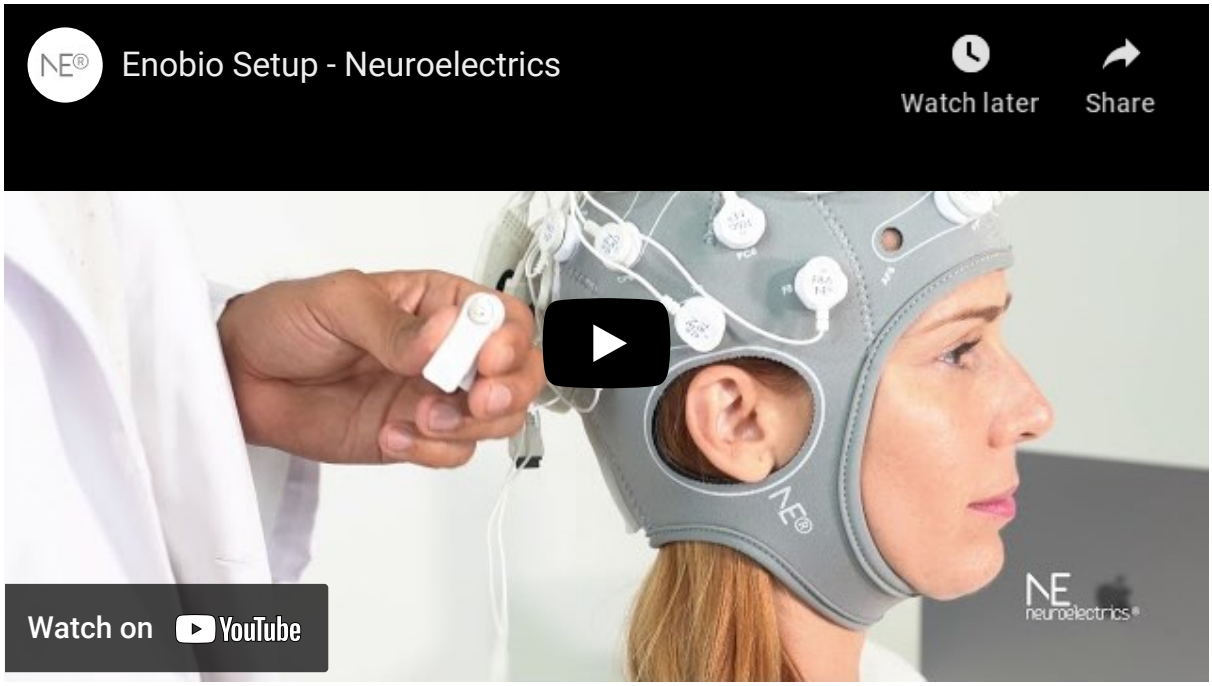
1. In iMotions, open Global Settings - Sensors tab, search for Lab Streaming Layer and device name.
2. Select ‘Enobio’ as Device Profile (note: LSL device profiles are only available for iMotions 9.1.6 or newer).
3. Click Search to find Outlets then select ‘LSLOutletStreamName-EEG’ (for EEG data from Enobio connections).
4. Check ‘EEG’ in Active Inlets.
5. Check ‘Enable data collection’
6. Click OK to save device settings and confirm that the device appears in your Sensor Preview widget and window.
7. Enable ‘Lab Streaming Layer’ in Study Sensors (when needed for data collection)



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Additional Information

- Enobio Setup video from NeuroElectrics (external link)
- NeuroElectrics website - external links: Download Area
- For more information on the NIC 2.0 software and troubleshooting the Enobio headset, refer to the Neuroelectrics Wiki: http://wiki.neuroelectrics.com.
- We recommend reviewing the Neuroelectrics FAQ section before conducting your study.

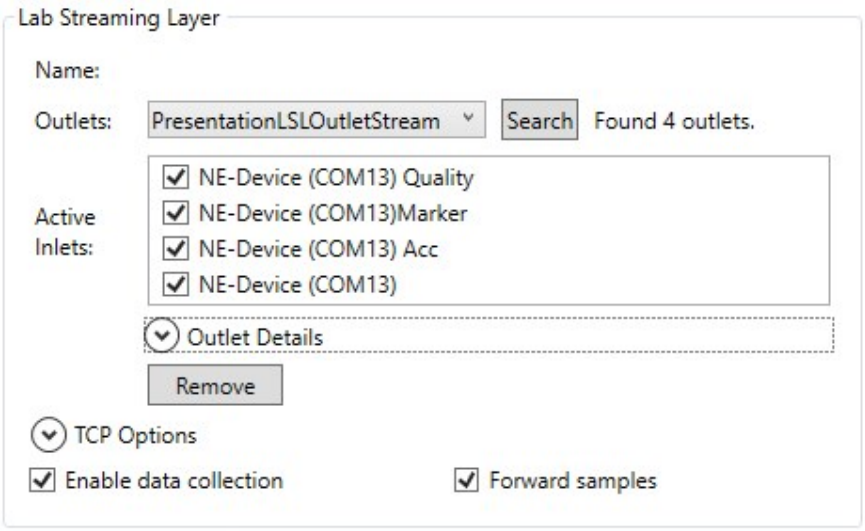


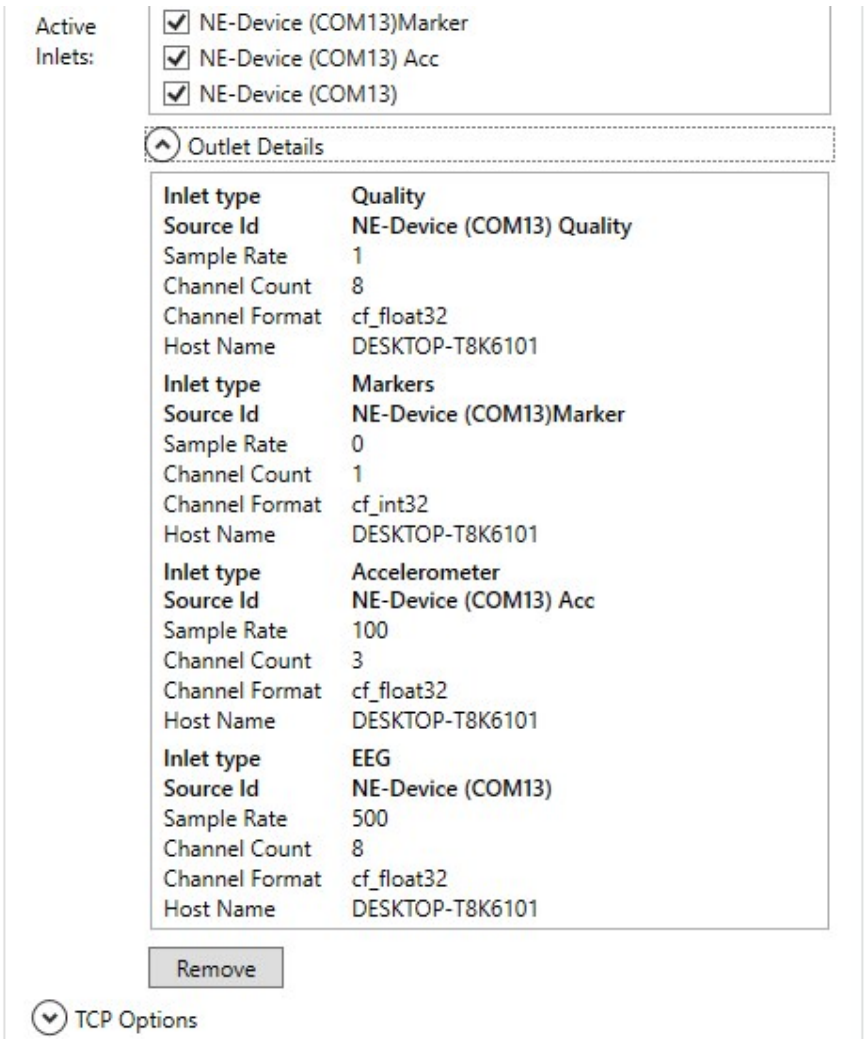
Enobio LSL Details

iMotions Global Settings can be used to check or adjust your LSL connection.

1. From iMotions main window, click Preferences > Sensor settings and locate the Lab Streaming Layer panel.
2. Search for outlets on Lab Streaming Layer. Enable/disable the channels as needed.
3. Refer to Outlet Details (drop-down menu) for additional LSL information (which received from your computer and the LSL source).

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